

Exploratory Data Analysis of Car Fuel Efficiency

A redesigned analysis of the Motor Trend *mtcars* dataset

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Course	BSc IT Hons
Project Name	Exploratory Data Analysis of Car Fuel Efficiency

Dataset: 32 automobile observations and 11 variables. The original project describes the dataset as Motor Trend road-test data from 1973–74 models.

1. Project Overview

This project examines the *mtcars* dataset to understand which vehicle characteristics are associated with fuel efficiency, measured in miles per gallon (mpg). The workflow covers data import, structure checks, missing-value checks, variable preparation, descriptive statistics, correlations and visualization.

Objectives

- Understand the dataset structure and variables.
- Check data quality and identify missing observations.
- Compare fuel efficiency across cylinder groups.
- Measure relationships between mpg and engine/vehicle characteristics.
- Present results using clear tables and charts.

Dataset Snapshot

Item	Value
Observations	32 cars
Variables	11 variables
Fuel-efficiency measure	mpg
MPG range	10.4 – 33.9
Average MPG	20.09
Missing values	0 in every column

The source project reports 32 observations and 11 variables with no missing values. Numeric codes such as cylinder count, transmission, gears and carburetors are interpreted as categorical variables where appropriate.

2. Data Preparation & Cleaning

The source workflow treats the built-in *mtcars* data as an externally supplied CSV: it exports the data, re-imports it with `read.csv()`, previews rows, checks structure, verifies missing values, and converts coded categories into readable labels.

Variables Used

Variable	Meaning	Role
mpg	Miles per gallon	Target
cyl	Number of cylinders	Category
disp	Engine displacement	Numeric
hp	Gross horsepower	Numeric
drat	Rear axle ratio	Numeric
wt	Vehicle weight (1000 lbs)	Numeric
qsec	Quarter-mile time	Numeric
vs	Engine shape	Category
am	Transmission type	Category
gear	Number of forward gears	Category
carb	Carburetors	Category

Cleaning Checks

Check	Result	Interpretation
Dataset dimensions	32 × 11	32 cars, 11 variables
Missing values	0	No imputation required
Cylinder coding	4 / 6 / 8	Converted to labeled groups
Transmission coding	0 / 1	Automatic / Manual labels

3. Statistical Exploration

Key descriptive statistics are reorganized below for a cleaner project presentation. These are the same source-supported measures used in the original analysis.

Variable	Minimum	Mean	Median	Maximum
MPG	10.40	20.09	19.20	33.90
Horsepower	52.00	146.69	123.00	335.00
Weight	1.51	3.22	3.33	5.42
Displacement	71.10	230.72	196.30	472.00
Quarter-mile time	14.50	17.85	17.71	22.90

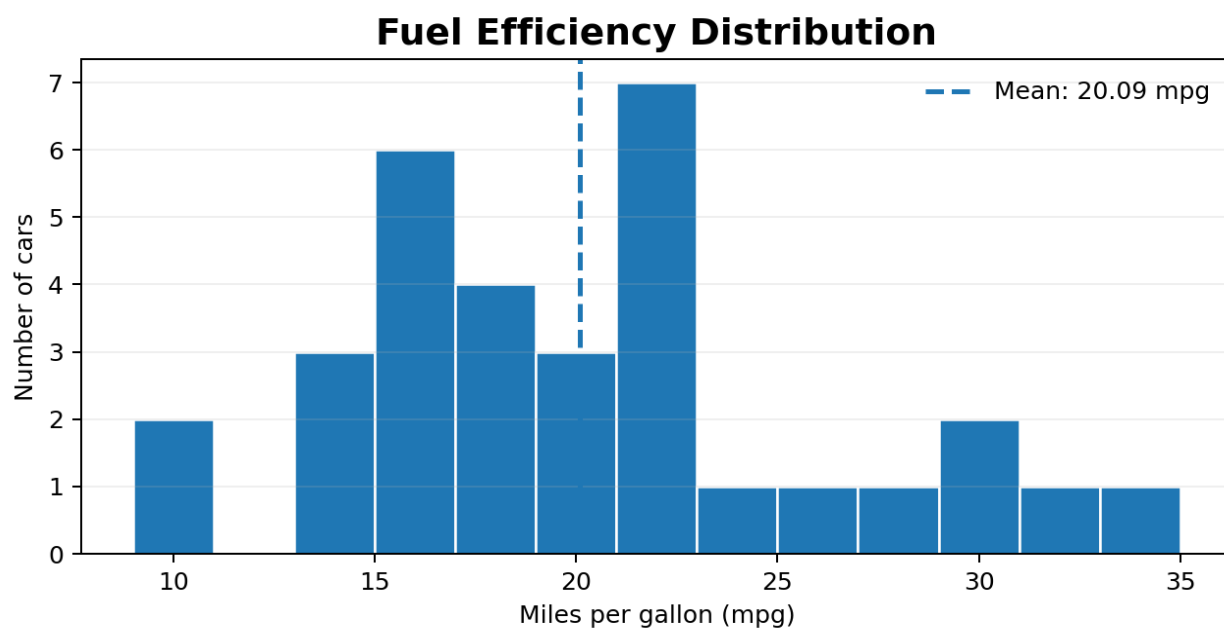
Cylinder Group Comparison

Cylinders	Cars	Average MPG	Average HP
4-cyl	11	26.66	82.6
6-cyl	7	19.74	122.3
8-cyl	14	15.10	209.2

The average fuel efficiency is 20.09 mpg. The source interpretation reports higher average mpg for cars with fewer cylinders and lower mpg as cylinder count increases.

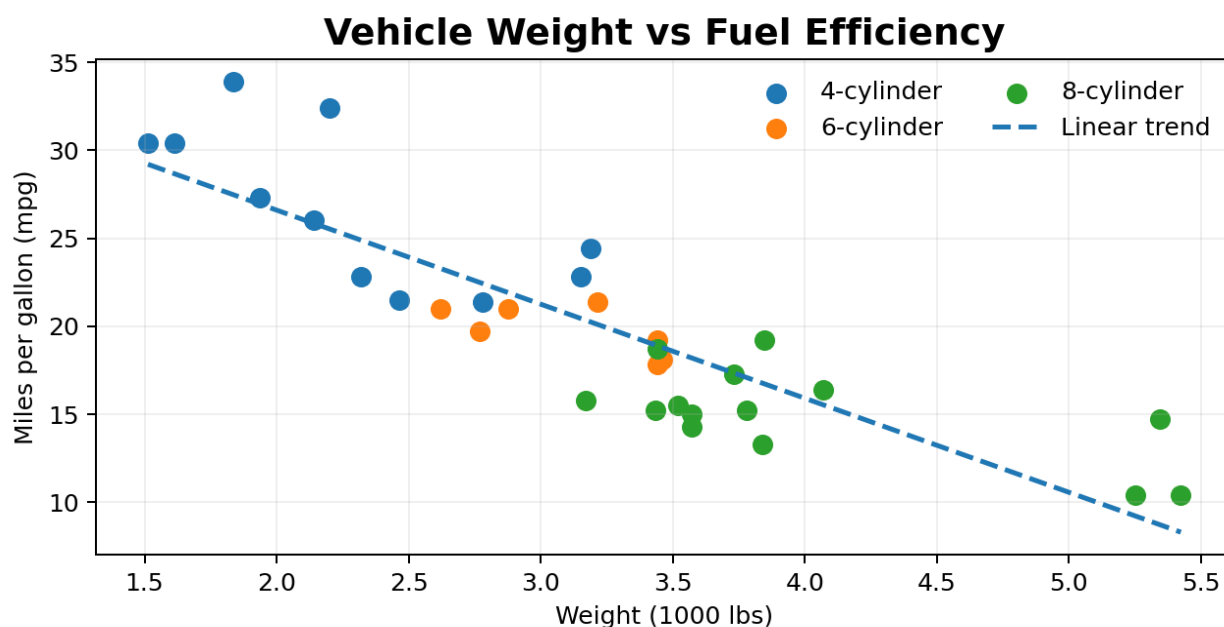
4. Visual Analysis

Fuel Efficiency Distribution



Most observations are concentrated in the middle of the mpg range, with a smaller number of highly efficient cars. The dashed line marks the overall mean.

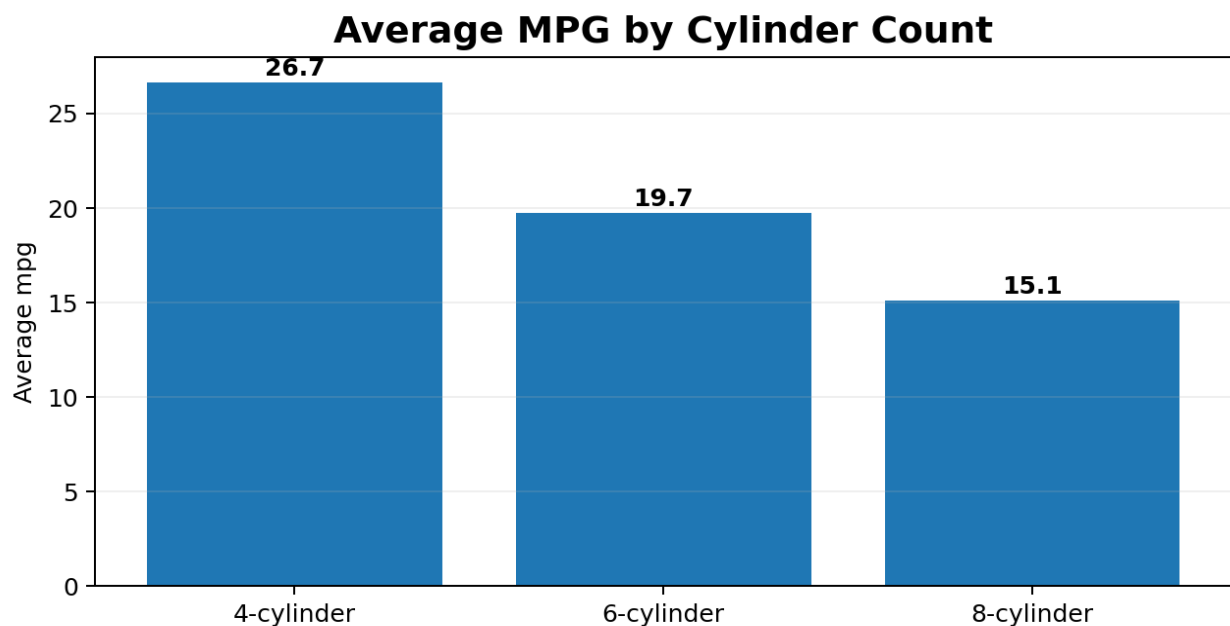
Weight vs Fuel Efficiency



The downward trend shows that higher vehicle weight is associated with lower mpg in this dataset. The source correlation is -0.87 for mpg and weight.

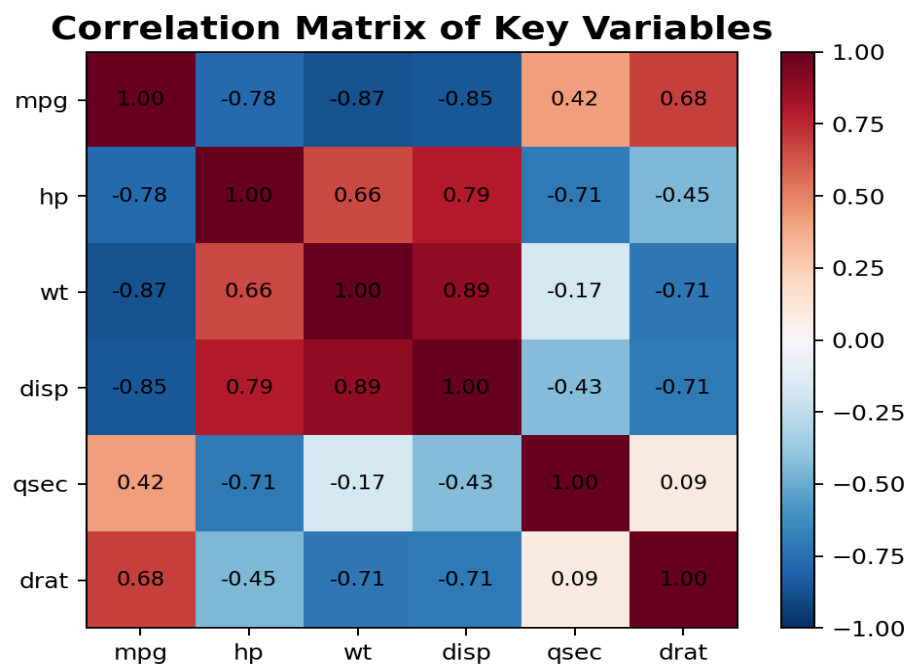
5. Group Comparison & Correlation

Average MPG by Cylinder Count



Average mpg decreases as cylinder count increases. This matches the source project's interpretation of the 4-, 6- and 8-cylinder groups.

Correlation Heatmap



The strongest negative relationships with mpg are weight (-0.87), displacement (-0.85) and horsepower (-0.78). Weight and displacement are strongly positively related (0.89).

6. Findings & Conclusion

Weight: The source analysis identifies weight as the strongest negative correlate of mpg among the key numeric variables, with a correlation of -0.87 .

Engine size: Displacement is strongly negatively correlated with mpg (-0.85), indicating lower fuel efficiency among larger-displacement cars in this sample.

Horsepower: Horsepower has a moderately strong negative correlation with mpg (-0.78).

Cylinders: Average mpg decreases from the 4-cylinder group to the 6-cylinder and 8-cylinder groups.

Data quality: All 32 observations have complete values for the 11 variables used in the project.

Conclusion

The exploratory analysis demonstrates a complete R-style workflow from data import and cleaning to statistical exploration and visualization. Within this 1973–74 vehicle dataset, fuel efficiency is most closely associated with vehicle weight, engine displacement, horsepower and cylinder count.

Important limitation

These relationships describe the 32 vehicles in the mtcars dataset. Correlation indicates association rather than proof of causation, and this dataset represents older vehicle models rather than the current vehicle market.

Project Summary

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